SP-COMPACT-1: Stable Identity and Dense Search State

Tactix Development Project

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Executive summary

`SP-COMPACT-1` is implemented and closed at commit `e1d35b8`. Tactix now keeps permanent external variable identities while compacting retired internal search indices. The implementation preserves the central architecture constraint: compaction is a formula-neutral kernel service, whereas the kissat-default 10% post-Reduce trigger belongs only to `DefaultPolicy`. No proof rule or guard depends on that trigger.

The final pin passes the full Release suite and focused sanitizer/assertion tests. Three same-window, no-throttle A/B tiers are certificate-clean. Tier A ties, Tier B improves by 23.25 PAR-2 and one solve, and Tier C is formally undecided: the aggregate loses 10.51 PAR-2 solely because one trajectory cliff flips the sign; without it, compaction improves by 68.99 PAR-2.

1 Implementation

The kernel owns bidirectional external–internal maps. DIMACS, proof, model, reconstruction, public API, and stored rule-instance literals retain their external names. Clauses, binaries, watches, arena references, trail and reasons, per-variable kernel state, and heuristic state use dense internal indices and move together.

The primitive refuses unsafe states and variables that have not completed formula retirement. Consequently it neither adds nor deletes a clause and has no proof delta. `DefaultPolicy` tests the strict kissat threshold after every Reduce; `TACTIX_NO_COMPACT=1` disables only that strategy choice. The final cursor fix stores raw external instance payloads, preventing renumbering from changing a macro-rule's resumption position.

2 Correctness evidence

Release CTest is green at 60/60. Focused assertion/address/undefined-sanitizer tests for kernel, scheduling, and solve-reset are green at 3/3. Leak detection was disabled because LeakSanitizer cannot start under the execution environment's ptrace wrapper; assertions and the address/undefined sanitizers remained active.

Tests exercise non-root trail and reasons, binary and large clauses, arena references, heuristic state, public rule instances, extension reconstruction, proof mapping, and stable scheduler keys. Three pre-final 10,000-case fuzz legs (DRAT, LRAT, and random-policy LRAT) found zero failures. Exact-final LRAT job `1594313` added 10,000 cases with zero failures and 9,827 verdicts. Its small formulas did not reach the threshold; real-instance probe `1594316` observed three compactions removing 2,796 internal variables. The modified fuzz-harness self-test passed in job `1594314`.

3 Performance evidence

The immutable binary SHA-256 is:

```
781ae152bf56a8860 788c2d29d39c72b4
bec84e4c5bf6de40 22c6dfd13e90cefef
```

Every window used an unthrottled SLURM array on `cpu-dense-hwulab-q` with 8 GB/core. Every returned Tactix proof and model verified.

Tier	Compaction	Disabled	Result
A	0.66, 199/200	0.66, 199/200	tie
B	69.63, 26/28	92.88, 25/28	+23.25 PAR-2
C	829.17, 22/30	818.65, 22/30	-10.51, one-mover fragile

The windows are A `20260808-221957-A-compact2`, B `20260808-222138-B-compact2`, and C `20260808-222726-C-compact2`. Tier B's sign survives removal of any five instances. Tier C's `mp1-Nb5T15` alone changes the aggregate: disabled solves SAT in 14.964 seconds while enabled times out at 1,200 seconds. Removing it flips the comparison to +68.99 PAR-2 for compaction. This is a trajectory cliff, not grounds for an instance-specific gate.

4 Decision and risks

Compaction ships enabled. It is the faithful kissat default, maintains the proof-rule/strategy separation, is correctness-clean, ties Tier A, and has a robust Tier B benefit. Tier C is explicitly recorded as undecided rather than recast as a win. The disabled arm remains available for diagnosis.

The first Tier C attempt, `20260808-221717-C-compact`, was cancelled after review found renumberable scheduler cursors; none of its numbers are used. Commit `e1d35b8` fixes that defect, and all cited windows use the corrected pin. Pre-existing untracked benchmark artifacts were not staged.

5 Next action

Continue the one-row-at-a-time Kissat-default parity campaign with `SP-VIVIFY-3`. Tier D shakedown, the full Tier D campaign, and the T3 gate remain downstream.